the exact expected result ρ_β, we can see the three stochastic methods perform excellently, whereas the results from using the Nosé-Hoover thermostat reveal extremely poor sampling as a result of the trajectory being trapped.

We would expect that using Nosé-Hoover Chains (8.16)–(8.21) may provide deterministic, ergodic sampling, by using a sufficiently long chain of artificial variables and correctly chosen mass parameters μ_i. However, the choice of these parameter values may prove crucial to the ergodicity of the dynamics. Therefore, we suggest that for equilibrium canonical sampling it is far more efficient to use a stochastic thermostat, in comparison to the ergodicity issues and additional parameter tuning required in the case of deterministic thermostats.

8.6 Isokinetic Methods and Integration with Very Large Timestep

In the last chapter we examined various numerical methods for Langevin dynamics, observing very substantial differences in the accuracy obtainable for configurational sampling using different schemes. In this section, we discuss methods that accelerate sampling using a more radical approach, namely modification of the ensemble to incorporate isokinetic constraints. We again assume in what follows that $\mathrm{N_c} = \mathrm{N_d}$, so there are no constraints acting on the system (other than the isokinetic constraint we are going to impose).

The starting point here is the Gaussian isokinetic thermostat [124]. This type of constraint can be introduced into Newtonian dynamics $\dot{\boldsymbol{q}} = \boldsymbol{M}^{-1}\boldsymbol{p}, \dot{\boldsymbol{p}} = \boldsymbol{F}(\boldsymbol{q})$ by adding a Lagrange multiplier, i.e. changing the equations of motion to

$$\begin{aligned}\dot{\boldsymbol{q}} &= \boldsymbol{M}^{-1}\boldsymbol{p},\\ \dot{\boldsymbol{p}} &= \boldsymbol{F}(\boldsymbol{q}) - \xi\boldsymbol{p},\\ K_0 &= K(\boldsymbol{p}) \stackrel{\text{def}}{=} \boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}.\end{aligned}$$

By differentiating the constraint and introducing the second equation, we obtain an equation for ξ:

$$0 = \boldsymbol{p}^T\boldsymbol{M}^{-1}\dot{\boldsymbol{p}} = \boldsymbol{p}^T\boldsymbol{M}^{-1}[\boldsymbol{F}(\boldsymbol{q}) - \xi\boldsymbol{p}].$$

This yields

$$\xi = \frac{\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{F}(\boldsymbol{q})}{\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}}.$$

Simply substitute this expression into the differential equations to obtain the unconstrained equations

$$\dot{\boldsymbol{q}} = \boldsymbol{M}^{-1}\boldsymbol{p}, \tag{8.37}$$

$$\dot{\boldsymbol{p}} = \boldsymbol{F}(\boldsymbol{q}) - \frac{\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{F}(\boldsymbol{q})}{\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}}\boldsymbol{p}. \tag{8.38}$$

These equations preserve $\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p} =$ constant. (Note: It is tempting to replace the denominator in this expression by K_0, the value of kinetic energy, but this is not recommended since it then destroys the first integral.) The system (8.37)–(8.38) can be rewritten in the alternative form

$$\dot{\boldsymbol{q}} = \boldsymbol{M}^{-1}\boldsymbol{p},$$
$$\dot{\boldsymbol{p}} = [I - \mathcal{H}]\,\boldsymbol{F}(\boldsymbol{q}),$$

where $\mathcal{H} = \boldsymbol{p}(\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p})^{-1}\boldsymbol{p}^T\boldsymbol{M}^{-1}$ is a *projector* matrix which is idempotent: $\mathcal{H}^2 = \mathcal{H}$. The invariant subspace associated to this projector is span$(\boldsymbol{p})$:

$$\mathcal{H}\boldsymbol{p} = \boldsymbol{p}.$$

Isokinetic dynamics has nonvanishing divergence, which requires a special treatment to elucidate its statistical mechanical properties [338, 376]. In order to find a stationary solution, one substitutes

$$\rho(\boldsymbol{q},\boldsymbol{p}) = e^{-w(\boldsymbol{q},\boldsymbol{p})}\Phi(K(\boldsymbol{p})), \tag{8.39}$$

(with Φ a nonvanishing function of K) into the continuity equation for the isokinetic system, the function w being at this point unknown. Then one obtains, with $\boldsymbol{f}_{\text{iso}}$ denoting the vector field in (8.37)–(8.38), that

$$\begin{aligned} 0 = \nabla\cdot(\rho\boldsymbol{f}_{\text{iso}}) &= (\nabla\rho)\cdot\boldsymbol{f}_{\text{iso}} + \rho(\nabla\cdot\boldsymbol{f}_{\text{iso}}) \\ &= (-\nabla w\rho + \rho\frac{\Phi'(K)}{\Phi(K)}\nabla K)\cdot\boldsymbol{f}_{\text{iso}} + \rho(\nabla\cdot\boldsymbol{f}_{\text{iso}}). \end{aligned}$$

Using the isokinetic property, the term involving K drops out. One next calculates the divergence of the isokinetic vector field and finds that (see Exercise 5)

$$\nabla\cdot\boldsymbol{f}_{\text{iso}} = (\mathrm{N_c} - 1)\frac{\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{F}}{\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}}.$$